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Your grade: 86.66%

Module 1 Summative Assessment

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Review Learning Objectives

Instructions

1 / 1 point

1. Which of the following is Bayes' rule?

☐

Ready to review what you've learned? Starting the assignment? I'm here to help.
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Help me practice

☐

$$P(A|B) = \frac{P(A|B)P(B)}{P(A)}$$

☐

$$P(A|B) = \frac{P(A|B)P(B)P(A)}{P(A)P(B)P(A)}$$

Due Oct 20, 11:59 PM EDT

Attempts 1 of 3 (3 attempts left)

Submit

Try again

☒

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Your grade: 86.66%

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

☒ Correct

Correct: $P(B|A)P(A) = P(A \cap B)$ and similarly $P(B|A)P(B) = P(A \cap B)$. From these facts, we can derive Bayes' rule.

2. You are at a party and there are two jars of candies. One jar contains 70% red candies and 30% blue candies, while the other jar contains 30% red candies and 70% blue candies. You randomly choose a jar but do not know which one you picked. What is the smallest number of candies you should randomly draw (with replacement) from the jar to guarantee your posterior probability is distinct from your prior?

☒ 1

☐ 3

☐ There is insufficient information

☐ 2

☒ Correct

Correct: Drawing a red candy will increase the posterior that you selected the first jar, while selecting a blue candy will decrease it.

3. Suppose a machine learning model is trained to classify emails as either spam or non-spam. The model has an accuracy of 90% in classifying non-spam emails correctly and an accuracy of 90% in classifying spam emails correctly. If 20% of all emails received are spam, what can Bayes' rule help us determine?

☒ The probability of an email being classified as spam given that it is actually spam.

☐ The probability of an email being classified as non-spam given that it is actually non-spam.

☐ The probability of an email being spam given that it is classified as spam.

☐ None of the above

☒ Incorrect

Bayes' rule is used to compute the probability of a hypothesis given some evidence. In this case, the hypothesis is that an email is spam, and the evidence is that the model classified the email as spam. Thus, Bayes' rule helps compute the probability of an email actually being spam (the hypothesis) given that it has been classified as spam (the evidence).

4. True or False: The posterior probability calculated using Bayes' Rule is always greater than the prior probability.

☒ False

☐ True

☒ Correct

Correct

5. Suppose you are attempting to determine which two distributions best describe a population. The distribution A is a uniform distribution between 0 and 1. The distribution B is a triangular distribution between 0 and 1 (a height of 3 at 0 and a height of 2 at 1). You have no reason to believe one is more likely than another. Data you collect yields the range from .25 to .75. What is the posterior probability of distribution B?

☒ 50%

☐ There is not enough information in this problem.

☐ 44.4%

☐ 62.5%

☒ Correct

Correct

6. Suppose the frequency of a disease in the population is 0.01. The true positive rate of a test is 95% and the true negative rate is 90% (meaning the false positive is 10%). What is the posterior probability of having the disease given "two positive test results"?

☐ 10%

☐ 90%

☒ 48%

☐ 22%

☒ Correct

Correct: The probability of having the disease rises significantly with two positive tests.

7. Suppose a pregnancy test has a false positive rate of 1% and a true positive rate of 99%. Also suppose an individual regularly uses contraception that is 99% effective every time they have sexual intercourse. Presume this to be your prior. After a full cycle with regular sexual activity, this individual receives a positive pregnancy test, what is the posterior probability of pregnancy?

☒ There is not enough information to compute this

☐ 75%

☐ 99%

☐ 90%

☒ Incorrect

While more information could potentially provide a more precise calculation, given the information provided, it is possible to compute the posterior probability using Bayes' rule and the provided prior information.

8. Suppose the frequency of a disease in the population is 0.01. The true positive rate of a test is 95% and the true negative rate is 90% (meaning the false positive is 10%). What is the lowest number of positive test results that will yield a posterior probability of having the disease higher than 95%?

☐ 1

☐ 3

☒ 4

☐ 2

☒ Correct

Four yields a posterior probability of over 95%.

9. Suppose you're trying to determine if a die is fair (meaning each face is equally likely), and you start with a 99% certainty that it's fair. What is the minimum number of times you need to roll a six in a row for your posterior probability that the die is fair to drop below 10%?

☒ 5

☐ There is insufficient information

☐ 7

☐ 6

☒ Correct

Correct: Considering all possible probabilities p for rolling a 6, we see that 5 of them is a row is so unlikely that our posterior dips below 10% despite starting with 99% certainty.

10. True or False: The likelihood is the probability of the data under the hypothesis.

☒ True

☐ False

☒ Correct

Correct

11. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes). You perform an experiment and flip the coin 4 times, all of which turn up tails. Suppose you believe there is a 50% chance the coin is fair, but your friend is more skeptical and begins with a prior of 50%. Which of you will see the biggest difference between your prior and your posterior?

☒ Your friend

☐ You

☐ You are tied

☐ It is impossible to tell

☒ Correct

Correct: Your friend starts with a more neutral prior (50%), meaning the data (4 tails) will have a larger impact on changing their posterior probability. This results in a bigger difference between their prior and posterior compared to yours.

12. Suppose you aim to spend less time on social media. You install an app that promises to reduce your social media time by 50% after the first week of using it; your total time spent on social media went down by 12%. What is your posterior probability about the app?

☐ 61%

☐ 73%

☒ There is not enough information

☐ 50%

☒ Correct

Correct: You do not know, for example, the standard deviation of your usage, is a 12% drop between two weeks part of your normal fluctuation?

13. The posterior will improve with more, randomly selected points of data.

☒ True

☐ False

☒ Correct

Correct: As more data is introduced, the sample provides a better representation of the distribution, and the posterior improves.

14. A court case is considering if an ICU nurse is guilty of murdering 4 of her patients. She was the only nurse who worked all four patients. The prosecution argues that the likelihood of the patients dying, assuming no foul play, is only 0.5%. Based on this, the jury deems her guilty. Were they correct to conclude this?

☐ Yes because the likelihood is low

☐ Yes because she was the only one on duty and the likelihood of her innocence is 0.5%

☐ No because the likelihood isn't low enough

☒ No because this is the wrong probability to consider

☒ Correct

Correct: The jurors are supposed to evaluate the probability that the nurse is innocent given the data. What they are given is the probability of the data given that the nurse is innocent. These are not the same at all.

15. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes). You believe the chance of it being fair is the same as if it is not. You perform an experiment and flip the coin 3 times, all of which turn up tails. Using Bayes' rule, what would you estimate to be the probability that the coin is fair given the data you have collected?

☐ 50%

☐ 46%

☒ 31.3%

☐ 13%

☒ Correct

Correct: The posterior probability ends up being 1/3.